Architecture Decision Records (ADR) – Two-Tier Web Service

1. Introduction:

Architecture Decision Records (ADRs) document important technical decisions, the reasons behind them, and their expected consequences. They help the development team understand why an architecture or technology was selected.

This document records the proposed architecture decisions for a two-tier web service. The decisions should be reviewed and validated before production deployment.

2. ADR-001: Select a Cloud Region:

Status: Proposed

Context:

The application needs a hosting region that provides acceptable response times, service availability, and compliance with applicable data requirements.

Decision:

Select a cloud region close to the primary users, subject to service availability, pricing, and data residency requirements.

Consequences:

- Reduces network latency for nearby users.
- May improve application responsiveness.
- Region availability and pricing must be checked.
- Disaster recovery requirements may require a secondary region.

3. ADR-002: Define Recovery Objectives:

Status: Proposed

Context:

Service interruptions and data loss can affect users and business operations. Recovery objectives must match the selected operating scenario.

Decision:

Use the following Recovery Time Objective (RTO) and Recovery Point Objective (RPO) targets from the planning worksheet.

Scenario	RTO	RPO
Lean	4 hours	24 hours
Balanced	1 hour	1 hour
Resilient	30 minutes	15 minutes

Consequences:

- Lean requires a less demanding recovery capability.
- Balanced requires more frequent backups and faster restoration.
- Resilient requires stronger recovery arrangements and more frequent data protection.
- Recovery tests are needed to verify whether the targets can be achieved.

RTO defines the target time to restore service. RPO defines the maximum targeted data-loss window.

4. ADR-003: Use Managed Secrets Storage:

Status: Proposed

Context:

Applications may require database credentials, API keys, and other sensitive configuration values. Hardcoding secrets in source code creates security risks.

Decision:

Use an appropriate managed secrets service when supported by the selected hosting platform. For environments without such a service, use secure configuration and access controls.

Consequences:

- Reduces exposure of credentials in source code.
- Supports controlled access and secret rotation.
- May add operational cost and configuration complexity.
- Requires appropriate permissions and secure deployment practices.

5. ADR-004: Apply Least-Privilege Access:

Status: Proposed

Context:

Excessive permissions can increase the impact of compromised accounts or application vulnerabilities.

Decision:

Grant users, services, and administrators only the permissions required to perform their assigned tasks.

Consequences:

- Reduces unnecessary access to application resources.
- Limits the potential impact of compromised credentials.
- Requires regular permission reviews.
- May require additional setup and testing.

6. ADR-005: Protect Backups and Test Recovery:

Status: Proposed

Context:

Backups are important for recovering from accidental deletion, corruption, infrastructure failures, and security incidents. A backup is useful only if the data can be restored successfully.

Decision:

Encrypt backups, restrict access, define retention periods, and test restoration procedures regularly. Configure backup frequency according to the selected scenario's RPO.

Consequences:

- Improves protection against data loss.
- Supports disaster recovery.
- Requires storage capacity and operational monitoring.
- Recovery tests help identify problems before an actual incident.

7. ADR-006: Monitor Service Reliability:

Status: Proposed

Context:

The service needs continuous visibility into availability, response time, errors, and resource usage.

Decision:

Monitor availability, p95 latency, error rates, application health, and database health. Configure alerts according to the selected scenario's SLO targets.

Scenario	Availability Target	p95 Latency Target
Lean	99%	800 ms
Balanced	99.5%	500 ms
Resilient	99.9%	300 ms

Consequences:

- Helps identify service degradation early.
- Supports incident investigation and performance improvements.
- Requires suitable monitoring tools and alert configuration.
- Monitoring costs should remain within the selected budget.

8. Review Process:

These decisions are proposed and should be reviewed when requirements, infrastructure, costs, or security needs change. Any significant change should be documented with its rationale and consequences.

9. Conclusion:

The ADRs establish a documented starting point for cloud region selection, recovery objectives, secret management, access control, backup protection, and service monitoring. They support consistent technical decisions and should be validated through implementation, cost review, and testing.